Student Performance Analytics Dashboard - Complete Beginner Guide

This project helps beginners learn complete Data Analytics workflow using Excel, MySQL, SQL, and Power BI.

Project Purpose

The purpose of this project is to analyze student academic performance and create interactive dashboards to identify top students, pass/fail trends, grade distribution, and subject-wise performance.

Software Downloads

- Microsoft Excel
- MySQL Community Server
- MySQL Workbench
- Power BI Desktop
- MySQL Connector NET
- Kaggle Student Dataset

Project Workflow

Dataset → Excel Cleaning → MySQL Database → SQL Analysis → Power BI Dashboard → Insights

Phase 1 - Dataset Collection

Download the dataset from Kaggle and extract the CSV file.

Phase 2 - Excel Data Cleaning

- Open dataset in Excel
- Check missing values
- Remove duplicate rows
- Rename columns using underscore format
- Create calculated columns
- Save file as CSV UTF-8

Excel Formulas Used

Total Marks: =F2+G2+H2

Average Marks: =AVERAGE(F2:H2)

Result: =IF(I2>=35,"Pass","Fail")

Grade: =IF(J2>=90,"A",IF(J2>=75,"B",IF(J2>=60,"C","D")))

Phase 3 - MySQL Database Setup

Create Database

```
CREATE DATABASE student_dashboard;
```

Use Database

```
USE student_dashboard;
```

Create Students Table

```
CREATE TABLE students (
    gender VARCHAR(20),
    race_ethnicity VARCHAR(50),
    parental_education VARCHAR(100),
    lunch VARCHAR(50),
    test_preparation VARCHAR(50),
    math_score INT,
    reading_score INT,
    writing_score INT,
    total_marks INT,
    average_marks DECIMAL(5,2),
    result VARCHAR(20),
    grade VARCHAR(5)
);
```

Import CSV into MySQL

Right click the students table → Table Data Import Wizard → Select CSV file → Finish

Phase 4 - SQL Analysis Queries

Total Students

```
SELECT COUNT(*) AS total_students FROM students;
```

Average Marks

```
SELECT AVG(average_marks) AS class_average FROM students;
```

Top 10 Students

```
SELECT * FROM students ORDER BY average_marks DESC LIMIT 10;
```

Pass vs Fail

```
SELECT result, COUNT(*) FROM students GROUP BY result;
```

Grade Distribution

```
SELECT grade, COUNT(*) FROM students GROUP BY grade;
```

Gender-wise Performance

```
SELECT gender, AVG(average_marks) FROM students GROUP BY gender;
```

Phase 5 - Power BI Dashboard

- Open Power BI Desktop
- Click Home → Get Data → MySQL Database
- Enter localhost and student_dashboard
- Login using MySQL username and password
- Load students table
- Create charts and KPI cardsPower BI Visuals Used

Detailed Explanation of Power BI Visuals Used

1. KPI Cards

Visual Used

Card Visual

Purpose

KPI (Key Performance Indicator) cards are used to display important summary values in a single number format.

These cards help users quickly understand overall performance without checking full tables.

KPI Cards Used in Project

KPI	Purpose
Total Students	Total number of students
Average Marks	Overall academic performance
Passed Students	Total students who passed

Fields Used

Total Students

Field	Aggregation
gender	Count

Why?

- Each row represents one student.
- Counting rows gives total students.

Average Marks

Field	Aggregation
average_marks	Average

Why?

- Average shows overall class performance.

Passed Students

Field	Aggregation
result	Count

Filter:

Pass

Why?

- Counts students whose result is “Pass”.

Why KPI Cards Are Important

They:
✅ provide quick insights
✅ improve dashboard readability
✅ summarize important information
✅ help decision-making faster

Interview Explanation

KPI cards are used to display important academic summary metrics like total students, average marks, and passed students for quick performance analysis.

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2. Pie Chart

Visual Used

Pie Chart

Purpose

Pie chart is used to show:

Pass vs Fail percentage distribution

It visually represents how many students passed and failed.

Fields Used

Section	Field
Legend	result
Values	result (Count)

How It Works

Power BI:

1. Groups data by:

Pass / Fail

2. Counts students in each category.
3. Displays percentages in circular format.

Why Pie Chart Was Selected

Pie charts are best for:
✅ percentage comparison
✅ part-to-whole analysis
✅ simple categorical data

Advantages

- Easy to understand

- Visually attractive
- Shows proportions clearly

Interview Explanation

Pie chart was used to analyze pass and fail distribution because it effectively visualizes percentage-based categorical comparisons.

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3. Clustered Column Chart

Visual Used

Clustered Column Chart

Purpose

Used for:

Gender-wise performance analysis

Compares average marks between:

- Male students
- Female students

Fields Used

Section	Field
X-axis	gender
Y-axis	average_marks

Aggregation: Average

How It Works

Power BI:

1. Groups students by gender.
2. Calculates average marks.
3. Displays vertical comparison bars.

Why Clustered Column Chart Was Selected

Best for:
✅ category comparison
✅ side-by-side analysis
✅ visual performance comparison

Advantages

- Easy comparison
- Clear visualization
- Professional appearance

Interview Explanation

Clustered column chart was used to compare average academic performance between male and female students.

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4. Bar Chart

Visual Used

Clustered Bar Chart

Purpose

Used for:

Grade Distribution Analysis

Shows how students are distributed across:

- Grade A
- Grade B
- Grade C
- Grade D

Fields Used

Section	Field
Y-axis	grade
X-axis	grade (Count)

How It Works

Power BI:

1. Groups students by grade.
2. Counts students in each grade.
3. Displays horizontal bars.

Why Bar Chart Was Selected

Bar charts are ideal for:
✅ categorical comparison
✅ frequency analysis
✅ ranking categories

Advantages

- Easy readability
- Good for category distribution
- Effective for comparing counts

Interview Explanation

Bar chart was used to visualize grade distribution because it clearly compares the number of students in each grade category.

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5. Table Visual

Visual Used

Table

Purpose

Used for:

Displaying top-performing students

Provides detailed row-level information.

Fields Used

Field
gender
average_marks
grade

Aggregation: Don't Summarize

How It Works

Displays actual student records in tabular format.

Sorted by:

average_marks DESC

Why Table Was Selected

Tables are best for:
✅ detailed information
✅ exact values
✅ row-level analysis

Advantages

- Detailed data visibility
- Supports sorting
- Supports filtering

Interview Explanation

Table visual was used to display detailed information of top-performing students including grades and average marks.

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6. Slicers

Visual Used

Slicer

Purpose

Slicers provide:

Interactive dashboard filtering

Users can dynamically filter all visuals.

Slicers Used

Slicer	Purpose
gender	Filter by gender
grade	Filter by grade
result	Filter by pass/fail

How It Works

When user selects:

Female

All dashboard charts update automatically.

Why Slicers Were Selected

Slicers are essential for:
✅ interactive dashboards
✅ dynamic analysis
✅ user-controlled filtering

Advantages

- Improves user experience
- Makes dashboard interactive
- Supports focused analysis

Interview Explanation

Slicers were used to provide interactive filtering functionality for analyzing data based on gender, grade, and result categories.

Overall Dashboard Purpose

The dashboard combines:

- KPI monitoring
- performance comparison
- category analysis
- detailed reporting
- interactive filtering

to create a complete student analytics system.

Final Dashboard Layout

Dashboard Title

KPI Cards

Pie Chart | Grade Chart | Gender Chart

Subject Comparison Chart

Top Students Table

Slicers / Filters

Dashboard Insights

📌 Total Students Analyzed: 1K
📌 Average Student Marks: 67.77
📌 Pass percentage is significantly higher than fail percentage
📌 Reading and writing scores are slightly higher than math scores
📌 Female students showed slightly higher average performance compared to male students

Advantages of the Project

- Beginner-friendly
- Resume-ready
- Uses industry tools
- Easy to explain in interviews
- Demonstrates SQL and Power BI skills

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